

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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*Dedicated to my family who endured missed birthdays,
missed festivals and countless days alone so that I may
make history*

0.1 Analysis Strategy

Background Modeling

The dominant and sub-dominant irreducible backgrounds for $H \rightarrow 4\ell$ process are ZZ production via $q\bar{q}$ annihilation and gg fusion respectively. The reducible background is of the instrumental nature and arises from the process where partial jets are misidentified as leptons. The main processes which contribute in this background are: jets production in association with Z boson, jets production in association with WZ boson pair and the $t\bar{t}$ pair production. From now on, this background will be denoted as $Z+X$.

TABLE 1: The number of observed candidates, expected background and signal events passing all the selections in the mass range of $105 < m_{4\ell} < 140$ GeV utilized in the $H \rightarrow 4\ell$ measurements. ZZ and signal events are estimated using MC simulations while $Z+X$ is estimated by control regions in data.

Channel	4e	4μ	$2e2\mu$
(8 TeV)			
$q\bar{q} \rightarrow ZZ$	3.05 ± 0.35	7.62 ± 0.49	9.03 ± 0.69
$Z + X$	1.52 ± 0.34	1.18 ± 0.48	4.16 ± 1.06
$gg \rightarrow ZZ$	0.20 ± 0.05	0.41 ± 0.10	0.50 ± 0.13
Total background expected	4.77 ± 0.71	9.21 ± 0.69	13.69 ± 1.29
$H \rightarrow 4\ell$ ($m_H = 125.0$ GeV)	2.93 ± 0.45	5.63 ± 0.67	7.29 ± 0.93
Observed	9	15	15
(7 TeV)			
$q\bar{q} \rightarrow ZZ$	0.84 ± 0.10	1.80 ± 0.11	2.24 ± 0.28
$Z + X$	0.34 ± 0.09	0.22 ± 0.09	1.03 ± 0.29
$gg \rightarrow ZZ$	0.03 ± 0.01	0.06 ± 0.02	0.07 ± 0.02
Total background expected	1.21 ± 0.14	2.08 ± 0.14	3.37 ± 0.40
$H \rightarrow 4\ell$ ($m_H = 125.0$ GeV)	0.70 ± 0.11	1.24 ± 0.14	1.67 ± 0.26
Observed	1	3	6

The irreducible backgrounds $q\bar{q} \rightarrow ZZ$ and $gg \rightarrow ZZ$ for $H \rightarrow 4\ell$ integrated and differential measurements are estimated from Monte Carlo (MC) simulation as done earlier in Ref. [Chatrchyan:2013mx]. In addition, following the study in Ref. [Bonvini:2013], in these measurements we have increased the expected yield for the $gg \rightarrow ZZ$ background by assigning to the leading order (LO) background cross section a $m_{4\ell}$ dependent k-factor equal to the one used for the signal $H \rightarrow 4\ell$ process. In the region ($105 < m_{4\ell} < 140$ GeV) this factor leads to an increase of the expected $gg \rightarrow ZZ$ background by a factor of ~ 2.66 . After the inclusion of this k-factor, the $gg \rightarrow ZZ$ process is still a sub-dominant background in all $H \rightarrow 4\ell$ measurements.

The reducible background ($Z+X$) is evaluated using a tight-to-loose “fake rate” method and the control regions in the data, following Ref. [Chatrchyan:2013mx].

The yields of the Z+X background are determined by two methods (opposite sign (OS) or same sign (SS)) are found to be consistent with each within statistical uncertainties. The first control region is obtained by requiring that two leptons passing loose selections which does fulfill the criteria of Z_1 candidate (failing ID and isolation requirements), while the other two leptons in the event passes the criteria of Z_1 candidate. The sample is represented as "2 Prompt + 2 Fail" (2P2F) sample. This sample is populated with the events contains only two prompt leptons (mostly Drell-Yan with small fraction of Z/γ^* and $t\bar{t}$ events). The second control region is obtained by requiring that three leptons pass the final requirements. The sample is represented as "3 Prompt + 1 Fail" (3P1F) sample.

The $m_{4\ell}$ shape of the reducible background is attained from OS method by performing a fit of $m_{4\ell}$ distributions of 3P1F and 2P2F events independently by empirical analytical functional constructed by Landau [landau:1944] in Ref. [Chatrchyan:2013mxa]. Prediction for the reducible background in all three channels together fitted using this empirical shape with indicated total uncertainty is shown in Figure 2. In case of the differential $H \rightarrow 4\ell$ measurements, the shape of the reducible background is obtained from the templates built from the control regions in data for each bin of the observable considered for the measurement, following a procedure similar to the one used in the spin-parity studies described in the Ref. [Chatrchyan:2013mxa]. More details on this procedure and template shapes for each of the differential $H \rightarrow 4\ell$ measurements will be given in Section ??.

The number of estimated signal and background events, along with observed data events passing all selections in the mass range used for $H \rightarrow 4\ell$ measurements ($105 < m_{4\ell} < 140$ GeV) are listed in Table 1 for 7 and 8 TeV.

Figure 1 shows the reconstructed four-leptons invariant mass distribution for whole 7 and 8 TeV data and MC samples in mass range ($50 < m_{4\ell} < 140$ GeV) covering both $H \rightarrow 4\ell$ and $Z \rightarrow 4\ell$ measurements range. Predictions for the reducible background in a wide range of $m_{4\ell}$ in all three channels together, fitted using the empirical shape with indicated total uncertainty, is separately shown in Figure 2.

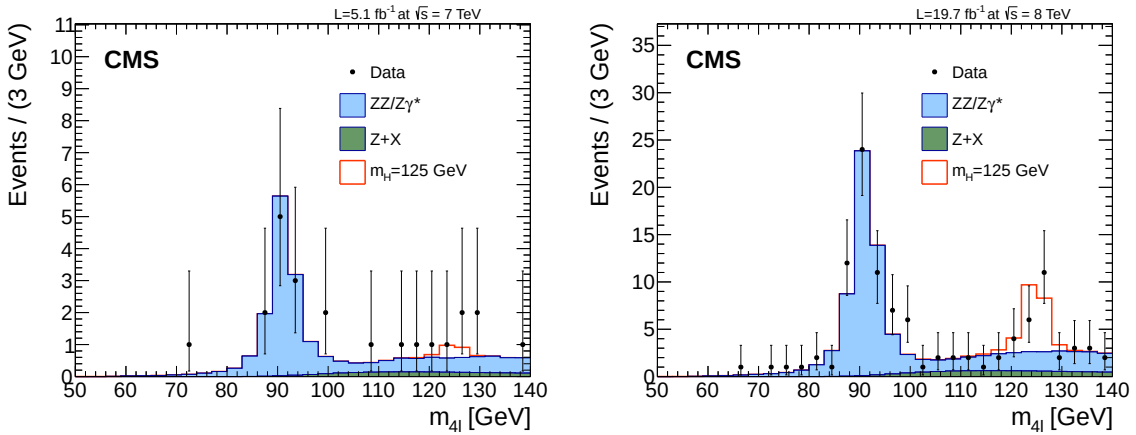


FIGURE 1: Distributions of invariant mass of four-leptons using 7 TeV (left) and 8 TeV (right) datasets, along with predictions for Standard Model Higgs ($m_H = 125.0$ GeV), including $Z \rightarrow 4\ell$ resonance.

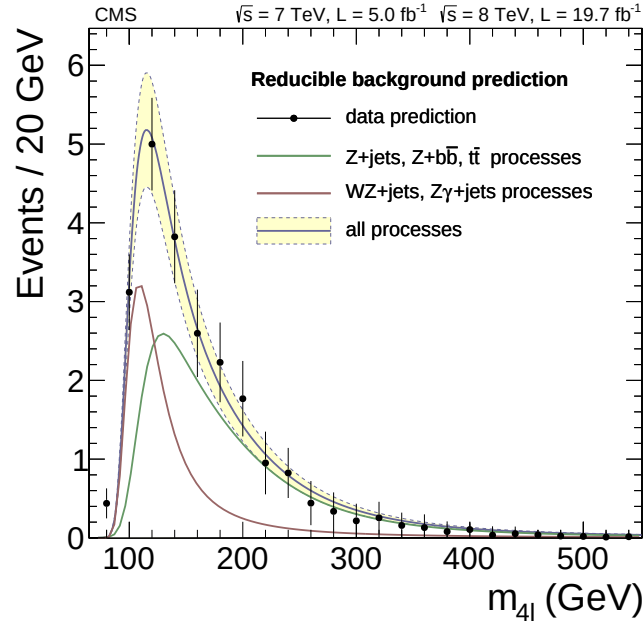


FIGURE 2: Prediction for the reducible background in all three channels together (black dots) fitted using an empirical shape (blue curve) with indicated total uncertainty (yellow band). Contributions from the 2P2F-like (green) and 3P1F-like (red) processes are fitted separately.